

ABBREVIATIONS

- VOLUME 2 IS NOT PHASED BUT INCLUDES THE NEW ADDITIONS TO LP-48.

SEE ELECTRICAL LEGEND

	SEE ELECTRICAL LEGEND
	
	
	
	
	
	
	
	

ELECTRICAL SWITCH/CALL BOX			FIRE HYDRANT
ELECTRICAL MANHOLE	— W _X ^H —	— W _H —	WATER LINE (HIGH FLOW SYSTEM)
TELECOMM MANHOLE	— W _X ^L —	— W _L —	WATER LINE (LOW FLOW / DOM SYSTEM)
ELECTRICAL HANDHOLE	— W _X ^{FP} —	— W _{FP} —	WATER LINE (FIRE PUMP SYSTEM)
TRANSFORMER	— W ₂₇ ¹⁰ —		WATER LINE (P1027 SYSTEM)
UNDERGROUND PRIMARY LINE	— W —	— W —	WATER LINE
UNDERGROUND SECONDARY LINE	— G _X —	— G —	UNDERGROUND GAS LINE
UNDERGROUND COMM LINE			GAS METER
OVERHEAD PRIMARY LINE			CONNECT TO EXISTING
			REINFORCE

G	NATURAL GAS	TOS	TOP OF SLOPE
GAL	GALLON(S)	TOS	TOP OF SLOPE
GBDAA	GROUND-BASED	TS	TEMPORARY SEEDING
	DETECT-AND-AVOID	TS	TOP OF STOOP
	GALVANIZED	TW	TOP OF WALL
GALV	GAS METER	TYP	TYPICAL
GM	GALLONS PER MINUTE	VERT	VERTICAL
GPM	GREASE TRAP	VOL	VOLUME
GT	HISTORIC, ARCHEOLOGICAL,	W	WEST/WATER
HAC	CULTURAL	W/	WITH
	HIGH DENSITY POLYETHYLENE	WM	WATER METER
HDPE	HANDHOLE	WV	WATER VALVE
HH	HORIZONTAL	WWF	WELDED WIRE FABRIC
HORIZ	HIGH POINT	XFMR	TRANSFORMER
HP		YD	YARD/YARD DRAIN
		YH	YARD HYDRANT

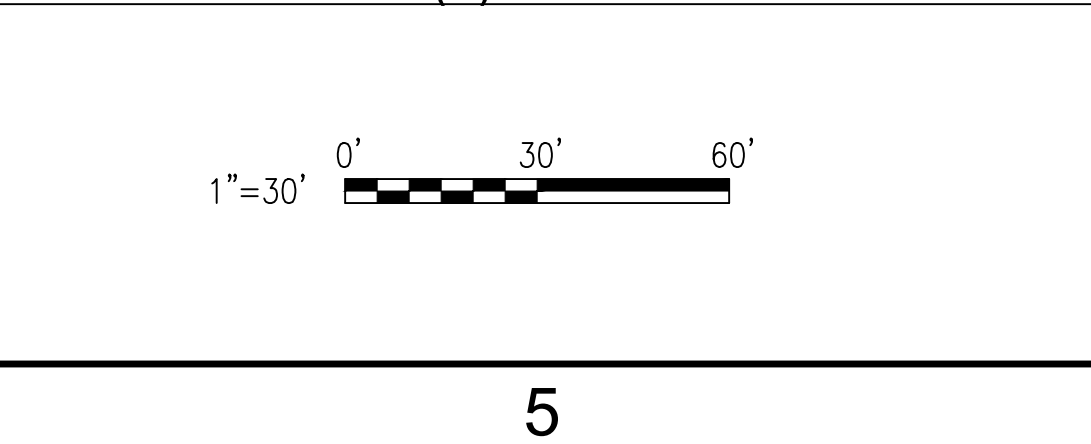
SCALE:	-
EPROJECT NO.:	1737012
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12924111
SHEET	32 OF 400
C-001	
DRAWFORM REVISION: 25 AUGUST 2020	



CONSTRUCTION NOTES

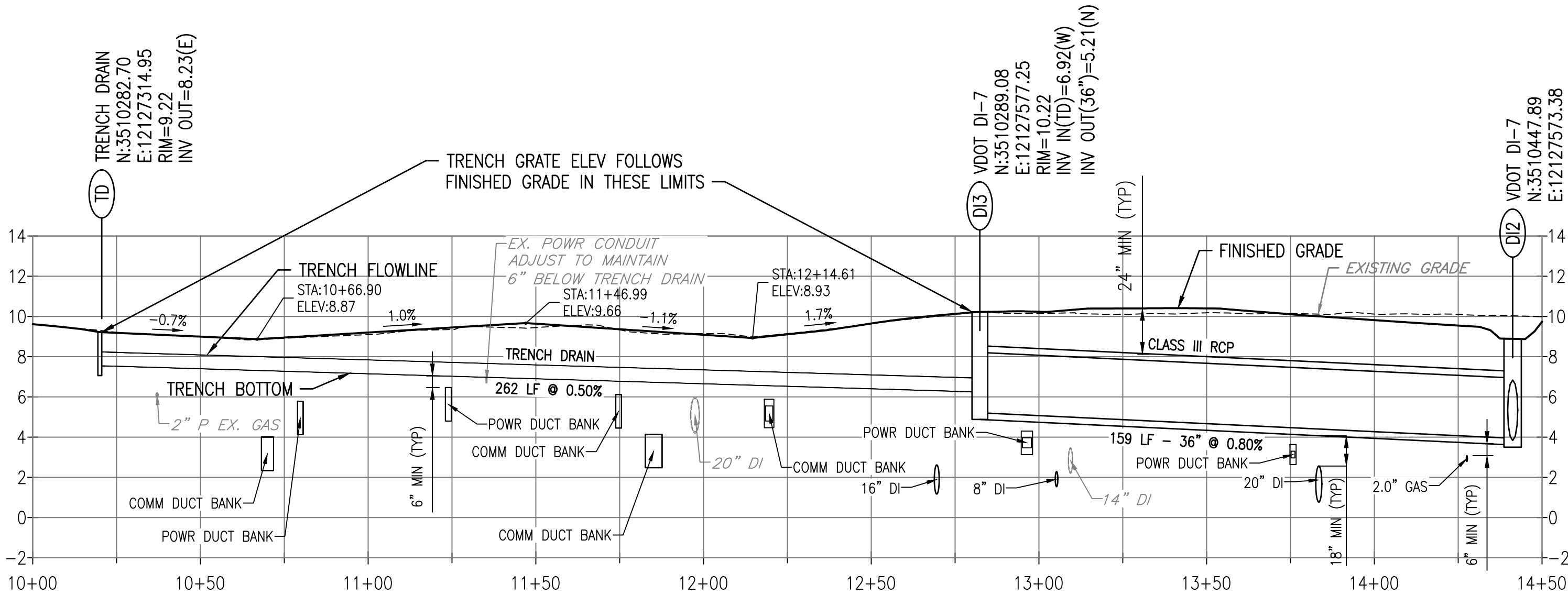
1. GUARD SHACK. GOVT REAL PROPERTY TEMPORARY GUARD SHACKS TO BE PROVIDED BY GOVT DURING CONSTRUCTION. REQ'D GUARD SHACKS TO BE PREFAB. GOVT REAL PROPERTY, AND INSTALLED BY CONTRACTOR. SEE A3/CS507 FOR MORE INFORMATION.
2. PEDESTRIAN TURNSTILE W/ABA GATES AND SALLYPORT. SEE CS507 FOR DETAIL.
3. WHEEL STOP. SEE C2/CS501 FOR DETAIL.
4. DUMPSTER ENCLOSURE WITH GATE AND BOLLARDS. SEE A3/CS502 FOR DETAIL.
5. 8" BOLLARD. SEE A1/CS502 FOR DETAIL.
6. TRANSFORMER PAD. SEE ELECTRICAL FOR DETAILS.
7. ARS POD STORAGE AREA. HATCHING IS NOT PAVEMENT MARKINGS AND IS FOR VISUAL CLARITY ONLY.
8. ACCESSIBLE PARKING. SEE A3/CS501 FOR DETAIL.
9. FLIGHT LINE SECURITY FENCE. SEE C1/CS503 FOR DETAIL.
10. DEADMAN AND CABLE REINFORCED FLIGHT LINE FENCE. SEE A2/CS505 FOR DETAIL. POSTS SHOWN FOR REFERENCE FOR REQUIRED DEADMAN CONFIGURATION.
11. 30' FLIGHT LINE INTERIOR CLEAR ZONE.
12. 33' ATFP BUILDING EXTERIOR UNOBSTRUCTED ZONE.
13. 20' FIRE LANE. TEXT MUST BE 18" HIGH, RETRO-REFLECTIVE YELLOW. MARKING MUST BE 6" WIDE, RETRO-REFLECTIVE YELLOW.
14. 4" THICK CONCRETE SLOPE PAVING BETWEEN FENCE AND DITCH.
15. ELEVATED SIDEWALK AND RAMP. SEE ARCHITECTURAL SHEETS FOR DETAILS.
16. SPLASH BLOCK. SEE C3/CS501 FOR DETAIL.
17. HEATED BFP ENCLOSURE. SEE CU101 AND CU502 FOR ADDITIONAL INFORMATION.
18. REPLACE DISTURBED FLIGHT LINE CLEAR ZONE PAVEMENT MARKING IN ACCORDANCE WITH NOTE 26.
19. REPLACE DISTURBED P1027 FIRE LANE PAVEMENT MARKING IN ACCORDANCE WITH NOTE 13.
20. GENERATOR PAD. SEE ELECTRICAL SHEETS FOR DETAILS.
21. MECHANICAL AND ELECTRICAL EQUIPMENT. SEE MECHANICAL, ELECTRICAL, AND STRUCTURAL SHEETS FOR INFORMATION.
22. DEADMAN CONFIGURATION. SEE NOTE 10 FOR INFORMATION.
23. SURFACE MOUNTED BOLLARD. SEE A2/CS503 FOR DETAIL.
24. 20' FLIGHT LINE EXTERIOR CLEAR ZONE.
25. STANDARD PARKING STALL PAVEMENT MARKINGS MUST BE 4" WIDE SOLID WHITE.
26. CLEAR ZONE AND EDGE OF DRIVE AISLE PAVEMENT MARKINGS MUST BE 4" WIDE SOLID YELLOW.
27. STOOP AND STEPS. SEE ARCHITECTURAL SHEETS FOR DETAILS.
28. DISTRIBUTION BOLLARD. SEE ELECTRICAL SHEETS FOR INFORMATION.
29. CROSSWALK PAVEMENT MARKING. SEE A2/CS501 FOR DETAIL.

GRAPHIC SCALE(S)

[illegible]

FILE NAME: I:\Projects\10000\10222-P-222 AC Laydown Fails FIA(BM-CAD)\10222 CAD-2A\Sheets\Volume 1\CG201-1.dwg LAYOUT NAME: CG201-1 PLOTTED: Monday, March 09, 2026 - 11:02am USER: wjdulley

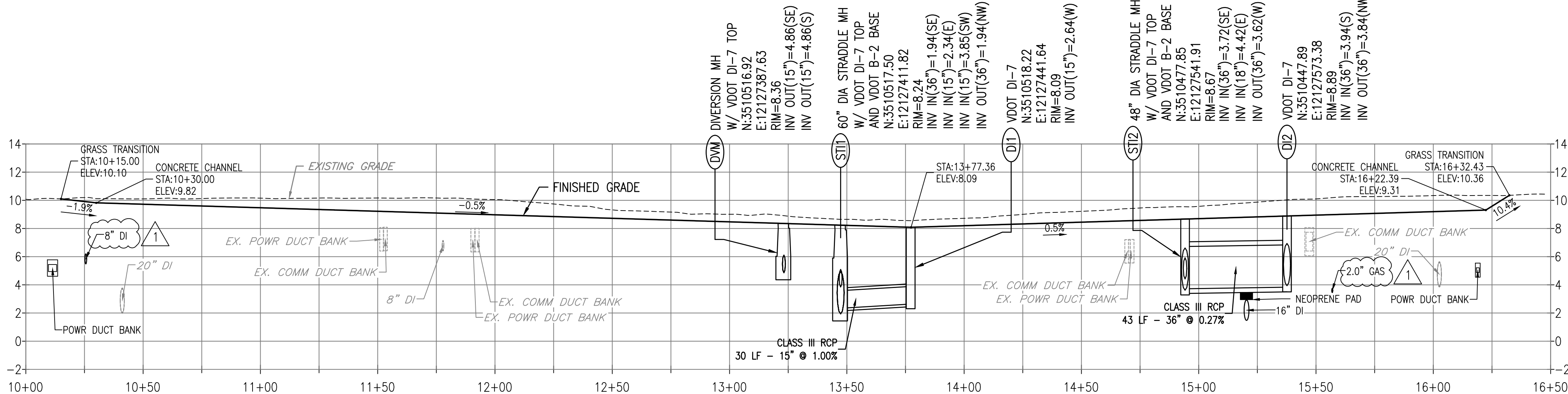
D



C1 TRENCH DRAIN SYSTEM PROFILE

HORIZ: 1"=30', VERT: 1"=5'

C



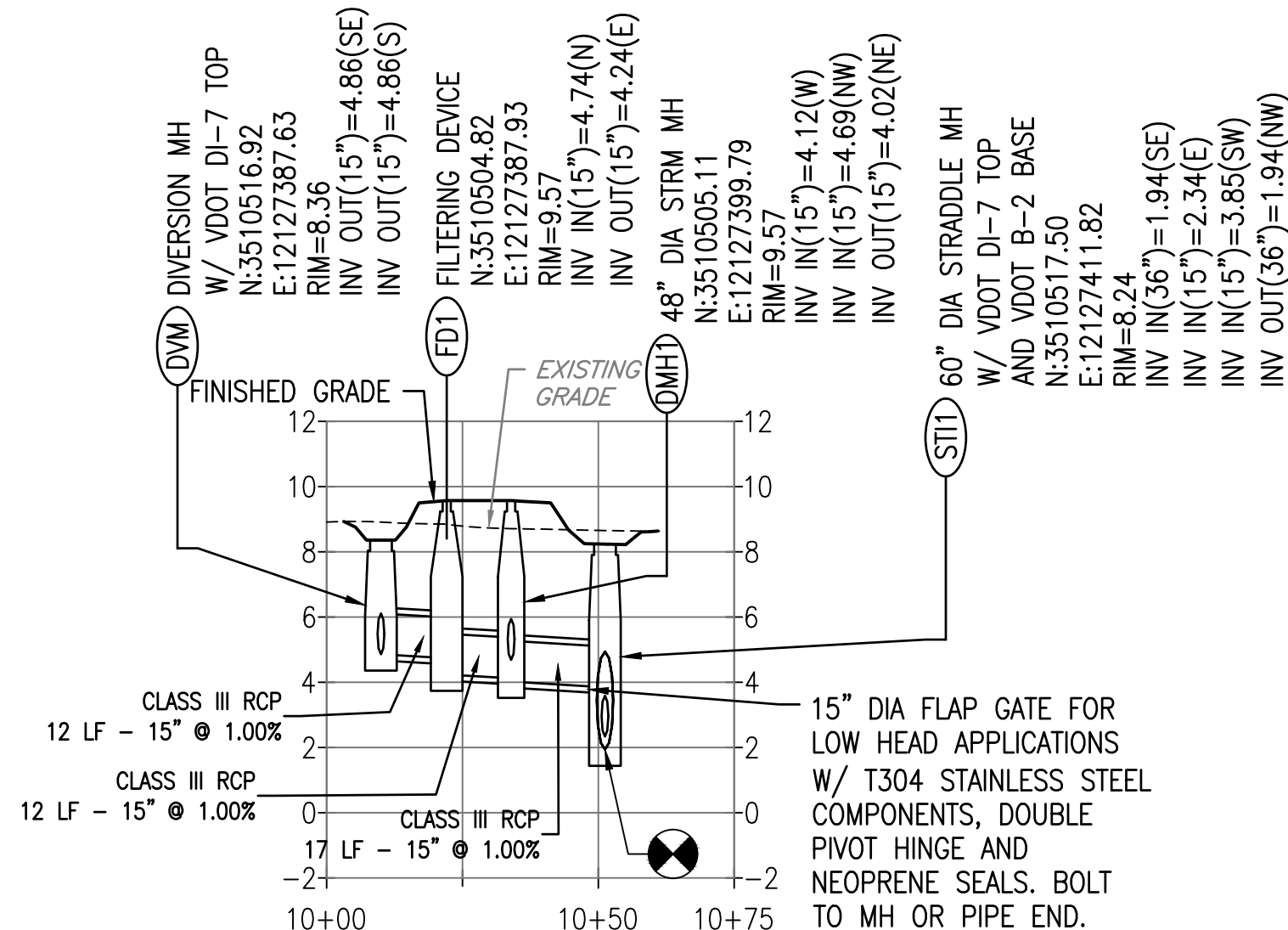
C4 STORM UPSIZE PROFILE

HORIZ: 1"=30', VERT: 1"=5'

B

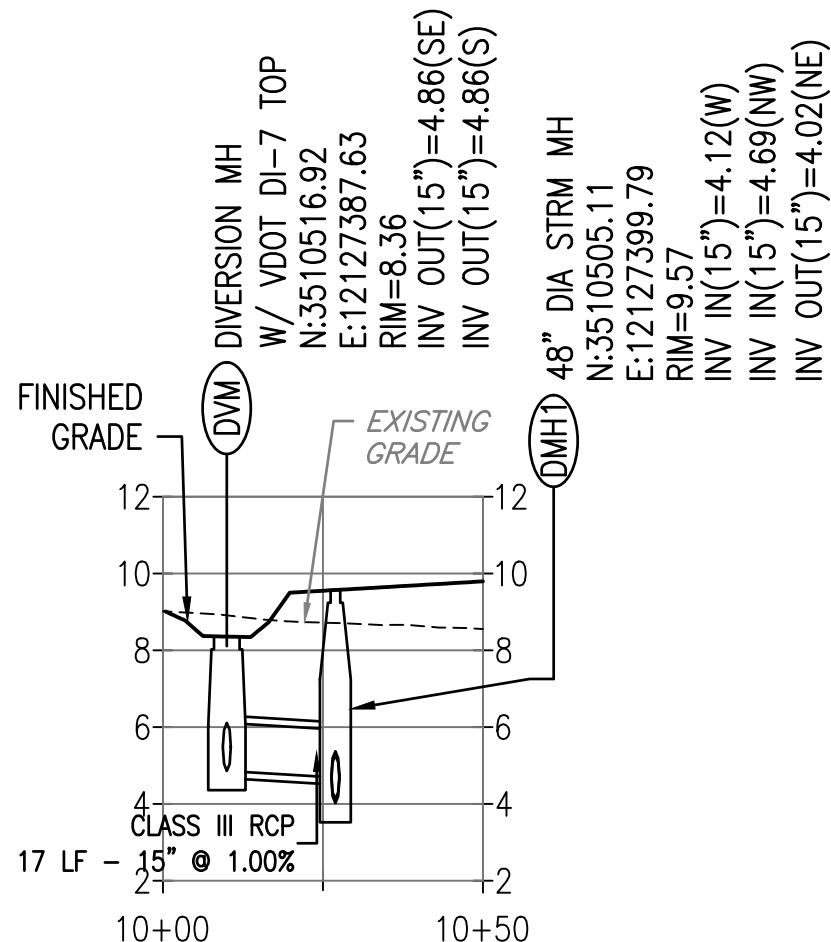
B1 CONCRETE DITCH PROFILE

HORIZ: 1"=30', VERT: 1"=5'



A1 TREATMENT PROFILE

HORIZ: 1"=30', VERT: 1"=5'



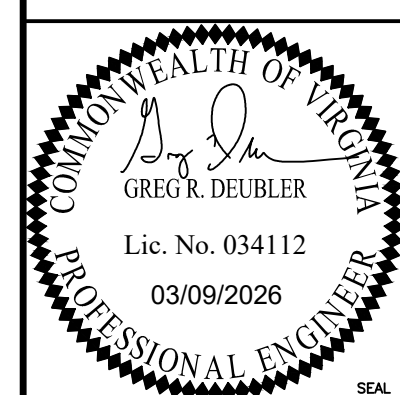
A2 OVERFLOW PROFILE

HORIZ: 1"=30', VERT: 1"=5'

A

GENERAL NOTES

1. WEEP HOLES ARE REQUIRED FOR ALL PROPOSED DRAINAGE STRUCTURES.
2. ALL PROPOSED STORM DRAINS AND PIPES MUST MAINTAIN AT A MINIMUM THE FOLLOWING VERTICAL CLEARANCES UNLESS OTHERWISE INDICATED:
 - 2.A. 6" FROM POWER AND COMM
 - 2.B. 18" FROM SANITARY AND WATER
 - 2.C. 6" FROM GAS AND OTHER STORM LINES
 - 2.D. 36" BELOW FINISHED GRADE
3. COORDINATE WITH ALL RELEVANT DISCIPLINES AND DISCIPLINES SHEETS TO PROVIDE A CONFLICT FREE CONSTRUCTION OF DRY AND WET UTILITIES AND STORM DRAINS AND PIPES AS INDICATED IN THE CONTRACT DOCUMENTS.



CLARKNEXSEN

4525 MAIN STREET, SUITE 1400
VIRGINIA BEACH, VA 23462
757.455.5800

APPROVED:

FOR COMMANDER NAVFAC 3/16/2026

ACTIVITY

Dave Snelling MQ-25 SEIT APML

SATISFACTORY TO DATE 05/28/2025

DES BSB DRW BSB CHK GRD/RLB

PMIDM RY / JPG

BRANCH MANAGER DLB

CHIEF ENGINEER JWW

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

P- 222 MQ25 AIRCRAFT LAYDOWN FACILITIES,

BUILDING LP-48A AND GBDAW TOWER

GRADING AND DRAINAGE PROFILES

SCALE: SCALE

EPROJCT NO.: 1737012

CONSTR. CONTR. NO.

NAVFAC DRAWING NO.

12924150

SHEET 71 OF 400

CG201

DRAWING REVISION: 25 AUGUST 2020

D

C

UNCLASSIFIED

B

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UNCLASSIFIED

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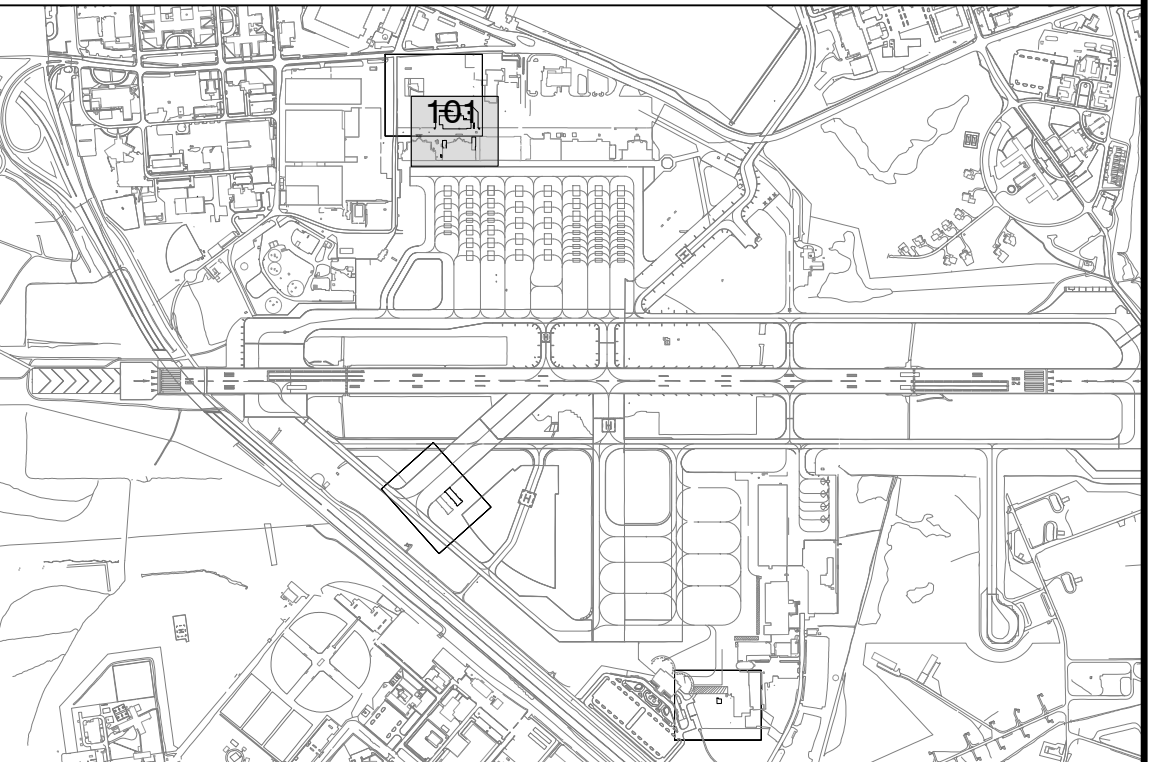
GENERAL NOTES

- COORDINATE WATER LINE CONNECTIONS WITH DEMO TERMINATION POINTS PROVIDED ON CD101.
- ALL PROPOSED HORIZONTAL BENDS INDICATED ARE 45° BENDS. REFER TO PROFILES PROVIDED ON CU201 AND CU202 FOR VERTICAL BEND LOCATIONS.
- FOR VALVE CONNECTIONS, USE MJ SWIVEL TEE WHERE CONNECTING TO TEE. USE BOLT-THROUGH POSITIVE RESTRAIN FITTINGS WHERE TEE NOT PROVIDED.
- REFER TO CU502 FOR PIPE RESTRAINT CHARTS.

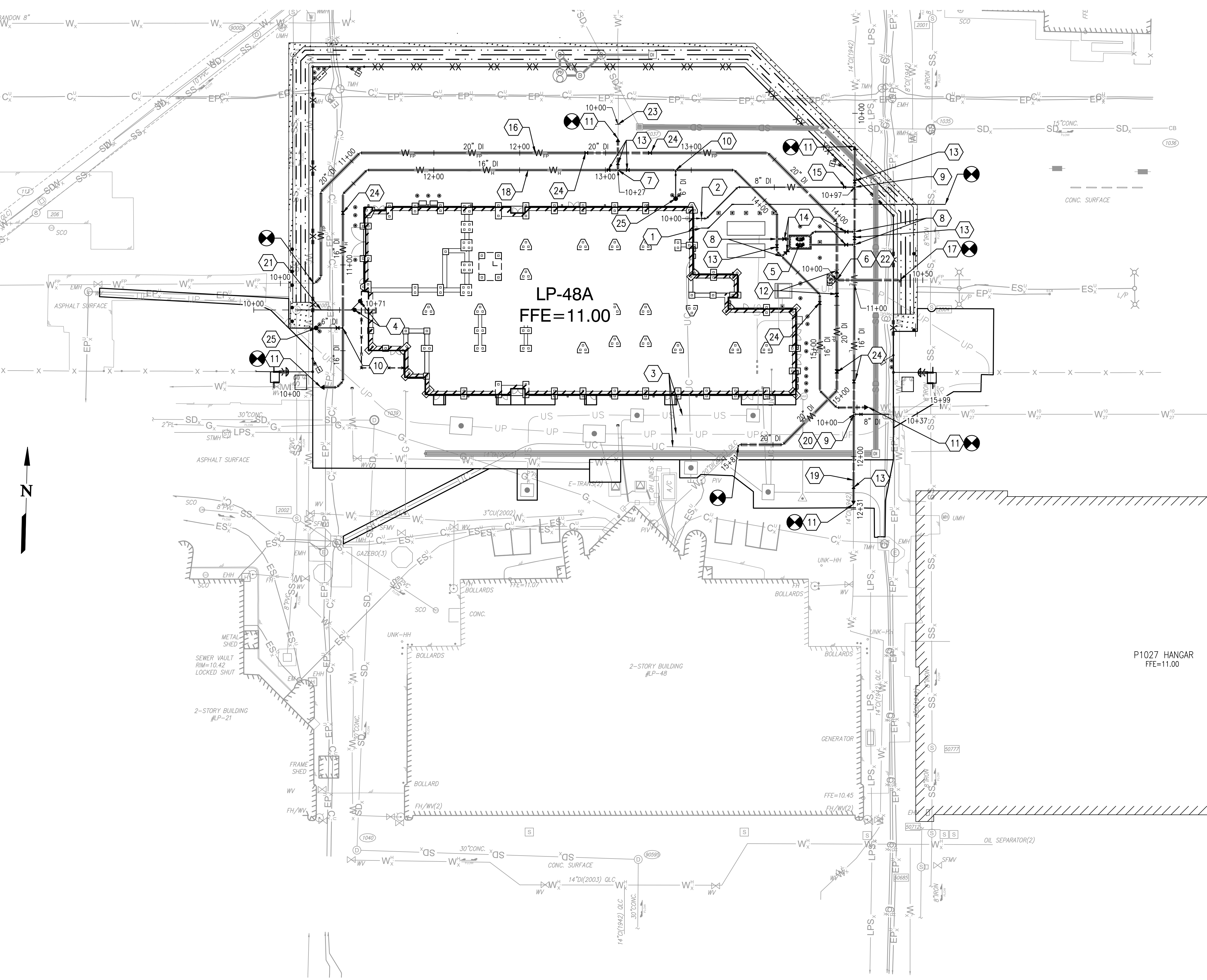
CONSTRUCTION NOTES

- 2.0" GAS LINE PROVIDED BY GAS COMPANY. SEE PLUMBING SHEETS FOR CONTINUATION. EXISTING GAS MAIN LOCATION NOT YET ESTABLISHED AND IS BEING INSTALLED FOR P1027.
- 8" DI FIRE AND DOMESTIC SERVICE. SEE C1/CU202 FOR PROFILE. SEE FIRE PROTECTION AND PLUMBING SHEETS FOR CONTINUATION. UNDERGROUND COMM AND UNDERGROUND POWER. SEE ELECTRICAL SHEETS FOR INFORMATION.
- SANITARY SEWER CLEANOUT. SEE PLUMBING FOR CONTINUATION. SEE C2/CU501 FOR DETAIL.
- TWO (2) RPZ DETECTOR TYPE BACKFLOW PREVENTERS AND HEATED DUAL ENCLOSURE. SEE A2/CU502 FOR DETAIL.
- 20"x20" TEE.
- 16"x16" TEE.
- 16"x12" TEE.
- 16"x8" TEE.
- 16"x6" MJ SWIVEL TEE.
- 16"x14" REDUCER.
- 20" GATE VALVE.
- 16" GATE VALVE.
- 12" GATE VALVE.
- 8" GATE VALVE.
- 20" DI LP33 FIRE PUMP SYSTEM LINE. SEE A1/CU201 FOR PROFILE.
- 20" DI LP33 FIRE PUMP SYSTEM LINE. SEE A1/CU202 FOR PROFILE.
- 16" DI HIGH FLOW SYSTEM LINE. SEE B1/CU201 FOR PROFILE.
- 16" DI LOW FLOW SYSTEM LINE. SEE C1/CU201 FOR PROFILE.
- 8" DI P1027 RECONNECTION LINE. SEE B1/CU202 FOR PROFILE.
- 8" DI SANITARY SEWER LATERAL. SEE C3/CU201 FOR PROFILE.
- THRUST BLOCK. SEE A1/CU502 FOR DETAIL.
- 16" DI HIGH FLOW SYSTEM NORTH RECONNECT LINE. SEE C2/CU202 FOR PROFILE.
- ARV. SEE C4/CU501 FOR DETAIL.
- FIRE HYDRANT AND VALVE ASSEMBLY. SEE A4/CU501 FOR DETAIL.

KEY PLAN



GRAPHIC SCALE(S)



A1 UTILITY PLAN - LP-48A
SCALE: 1" = 30'

DATE	03/11/26	APPR
SYN	DESCRIPTION	
1	MISC. REVISIONS	

4525 MAIN STREET, SUITE 1400
VIRGINIA BEACH, VA 23462
757.455.5800

APPROVED:
FOR COMMANDER NAVFAC ACTIVITY: Dave Snelling MQ-25 SEIT APML

SATISFACTORY TO	DATE	05/28/2025			
DES	SSB	DRW	SSB	CHK	GRD/RLB
PMIDM	RY / JPG				
BRANCH MANAGER	DLB				
CHIEF ENGINEER	JWU				
FIRE PROTECTION	HPN				

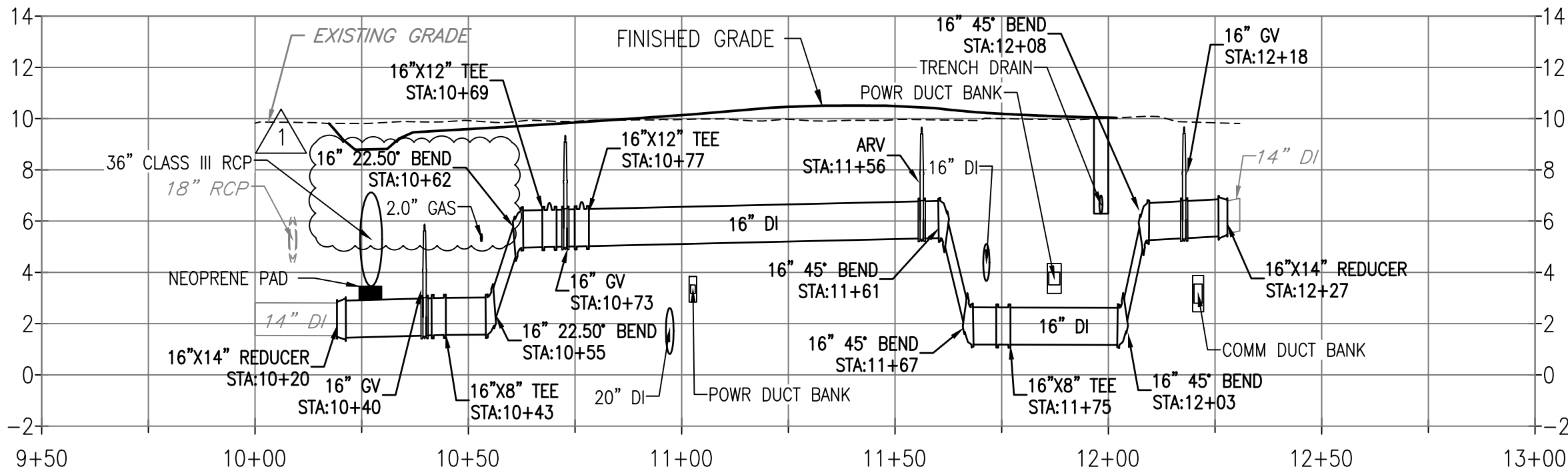
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL STATION - NORFOLK, VA
P- 222 MQ25 AIRCRAFT LAYDOWN FACILITIES,
BUILDING LP-48A AND GBDA TOWER
UTILITY PLAN - LP-48A

SCALE:	SCALE
PROJECT NO.:	1737012
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12924161
SHEET	82 OF 400
CU101	

DRAWING REVISION: 25 AUGUST 2020

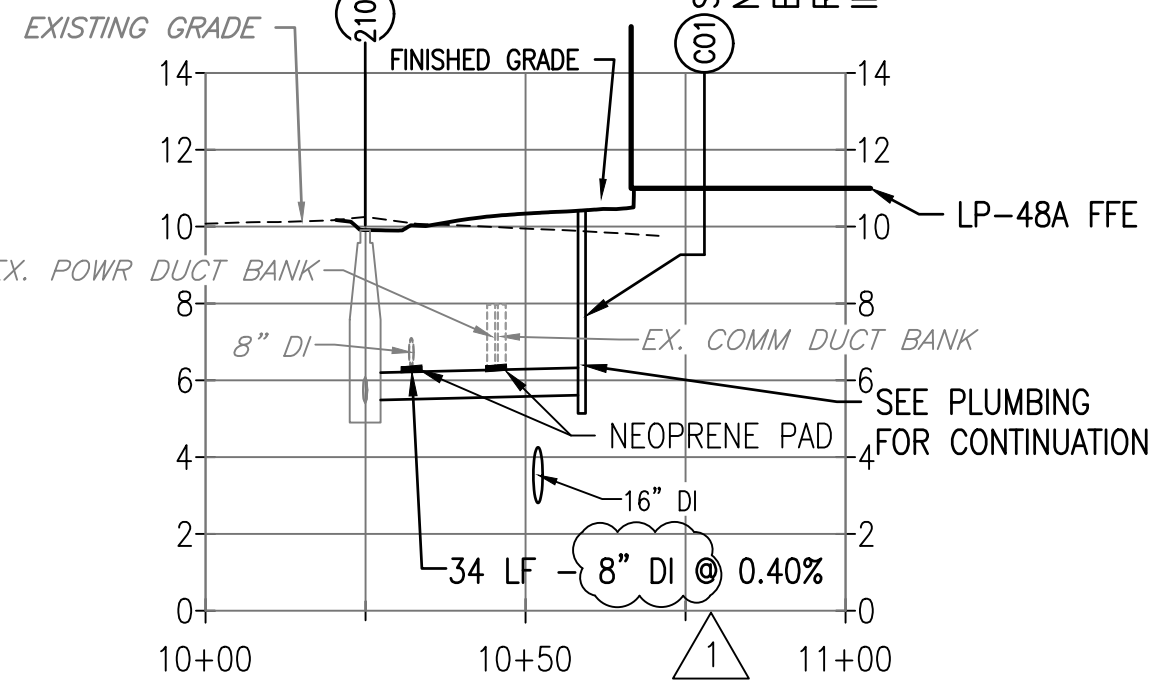
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D



C1 DOMESTIC SYSTEM

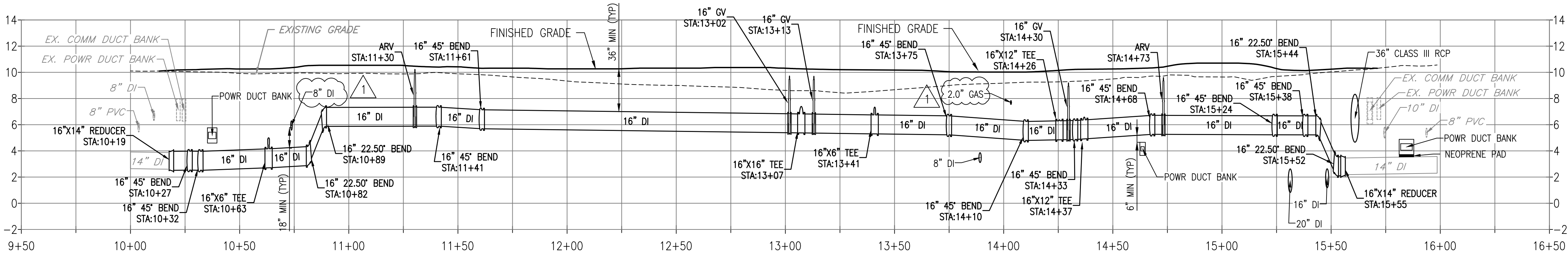
HORIZ: 1"=30', VERT: 1"=5'



C3 LP-48A SANITARY SEWER LATERAL

HORIZ: 1"=30', VERT: 1"=5'

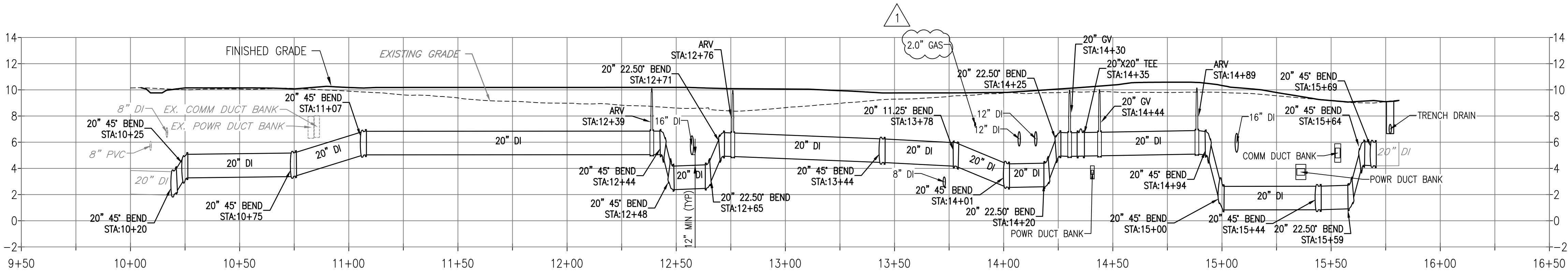
C



B1 HIGH FLOW SYSTEM

HORIZ: 1"=30', VERT: 1"=5'

B



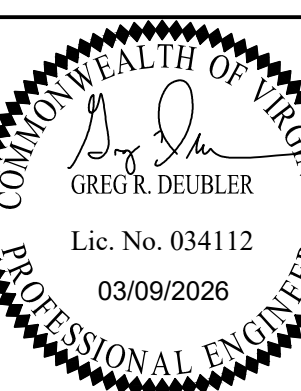
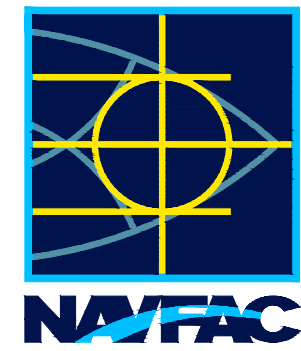
A1 LP33 FP SYSTEM WEST & LP48 RECONNECTION

HORIZ: 1"=30', VERT: 1"=5'

A

GENERAL NOTES

1. ALL PROPOSED WATER LINES MUST BE MECHANICALLY RESTRAINED IN ACCORDANCE WITH THE SPECIFICATIONS AND AS INDICATED ON CU502.
2. ALL PROPOSED WATER LINES MUST MAINTAIN AT A MINIMUM THE FOLLOWING VERTICAL CLEARANCES UNLESS OTHERWISE SHOWN AT SHALLOW UTILITY CROSSINGS, INDICATED BY NEOPRENE PADDING:
 - 2.A. 6" FROM POWER AND COMM (COORDINATE LOCATIONS WITH ELECTRICAL AND COMM SHEETS)
 - 2.B. 18" FROM STORM AND SANITARY
 - 2.C. 12" FROM GAS AND OTHER WATER LINES
 - 2.D. 36" BELOW FINISHED GRADE
3. COORDINATE WITH ALL RELEVANT DISCIPLINES AND DISCIPLINES SHEETS TO PROVIDE A CONFLICT FREE CONSTRUCTION OF DRY AND WET UTILITIES AND STORM DRAINS AND PIPES AS INDICATED IN THE CONTRACT DOCUMENTS.



CLARKNEXSEN

4525 MAIN STREET, SUITE 1400
VIRGINIA BEACH, VA 23462
757.455.5800

APPROVED
FOR COMMANDER NAVFAC

ACTIVITY

Dave Snelling MQ-25 SEIT APML

SATISFACTORY TO DATE 05/28/2025

DES BSB DRW BSB CHK GRD/RLB

PMIDM RY / JPG

BRANCH MANAGER DLB

CHIEF ENGINEER JWW

FIRE PROTECTION HPN

NAVFACILITIES ENGINEERING SYSTEMS COMMAND

NAVFACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

P- 222 MQ25 AIRCRAFT LAYDOWN FACILITIES,

BUILDING LP-48A AND GBDA TOWER

UTILITY PROFILES

SCALE: SCALE

EPROJCT NO.: 1737012

CONSTR. CONTR. NO.

NAVFAC DRAWING NO.

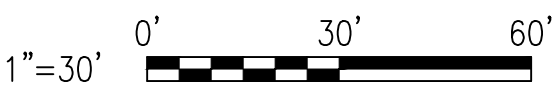
12924162

SHEET 83 OF 400

CU201

DRAWING REVISION: 25 AUGUST 2020

GRAPHIC SCALE(S)



D

C

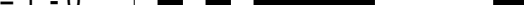
UNCLASSIFIED

B

A



1/8" = 1' - 0"



0 5' 10' 15' 30'

[illegible]



A1 FIRST FLOOR PLAN - AREA B
1/8" = 1'-0"

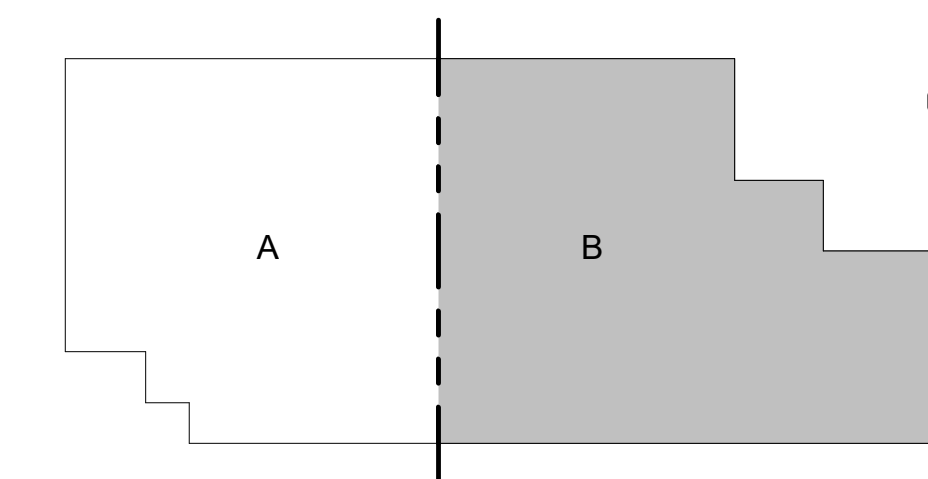
GENERAL NOTES

- REFER TO OVERALL FLOOR PLANS FOR ADDITIONAL GENERAL COMMENTS PERTINENT TO THIS SHEET.
- REFER TO DOOR SCHEDULE ON AE601 FOR ADDITIONAL DOOR INFORMATION.

KEY NOTES

- WALL MOUNTED DISPLAY MONITOR, COORDINATE WITH AUDIOVISUAL DRAWINGS
- WALL MOUNTED PORCELAIN MARKERBOARD, 60"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- WALL MOUNTED PORCELAIN MARKERBOARD, 72"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- WALL MOUNTED PORCELAIN MARKERBOARD, 84"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- ELECTRICAL EQUIPMENT, COORDINATE WITH ELECTRICAL DRAWINGS
- MECHANICAL EQUIPMENT, COORDINATE WITH MECHANICAL DRAWINGS
- BOLLARD, COORDINATE WITH CIVIL DRAWINGS
- VIRTUAL REALITY TRAINER EXTENT (NIC)
- ELECTRIC TIME ZONE CLOCK ABOVE
- RACK STORAGE LOCATION, REFER TO AUDIOVISUAL DRAWINGS
- DISPLAY CONTROL LOCATION, REFER TO AUDIOVISUAL DRAWINGS
- DOOR STOOP AND RAMP, SEE A1/AE414 FOR GUARDRAIL DETAIL, REFER TO CIVIL DRAWINGS FOR RAMP DETAILS
- WALL MOUNTED TACKBOARD, 48"W X 36"H, MOUNTED AT 5'-6" AFF TO CENTER, COORDINATE WITH INTERIORS DRAWINGS

KEY PLAN



GRAPHIC SCALE(S)



APPR	DATE	3/1/26
SYM	DESCRIPTION	1. MISS. REVISIONS
CLARK NEXSEN		
4525 MAIN STREET, SUITE 1400 VIRGINIA BEACH, VA 23462 757.455.5800		
APPROVED:		
FOR COMMANDER NAVFAC		
ACTIVITY: Dave Snelling MQ-25 SEIT APML		
SATISFACTORY TO DATE: 05/28/2025		
DES	RAM	DRW
KCM	CHK	SEB
PMIDM	RY / JPG	
BRANCH MANAGER	DJD	
CHIEF ENGINEER	JWU	
FIRE PROTECTION	HPN	
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND ~ MID-ATLANTIC NAVAL STATION - NORFOLK, VA P-222 MQ-25 AIRCRAFT LAYDOWN FACILITIES, BUILDING LP-48A AND GBDA TOWER FIRST FLOOR PLAN - AREA B		
SCALE: AS NOTED		
EPROJCT NO.: 1737012		
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO. 12924204		
SHEET 125 OF 400		
AE101B		
DRAWING REVISION: 26 AUGUST 2020		



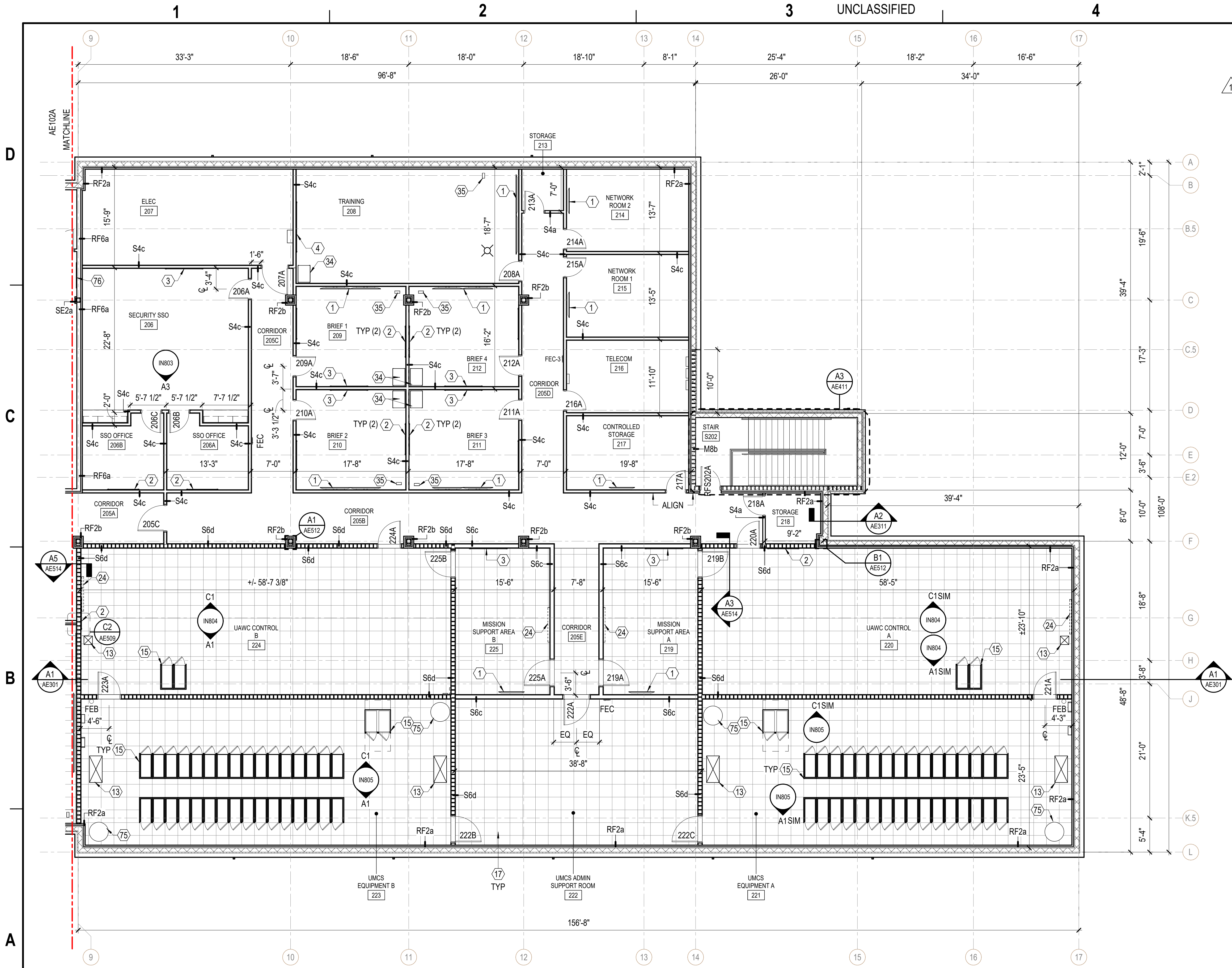
1/8" = 1' - 0"



0 5' 10' 15' 30'

DRAWFORM REVISION: 25 AUGUST 2020

A



A1 SECOND FLOOR PLAN - AREA B
1/8" = 1'-0"

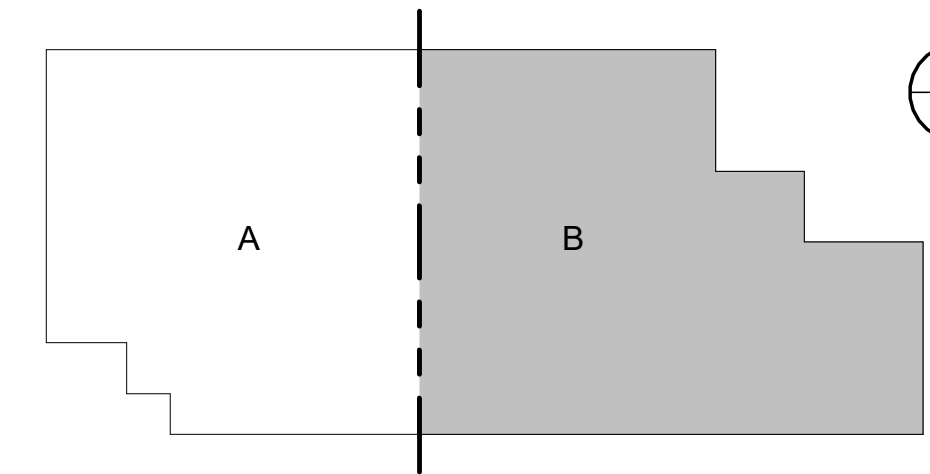
GENERAL NOTES

- A. REFER TO OVERALL FLOOR PLANS FOR ADDITIONAL GENERAL COMMENTS PERTINENT TO THIS SHEET.
- B. REFER TO DOOR SCHEDULE ON AE601 FOR ADDITIONAL DOOR INFORMATION.

KEY NOTES

- 1 WALL MOUNTED DISPLAY MONITOR, COORDINATE WITH AUDIOVISUAL DRAWINGS
- 2 WALL MOUNTED PORCELAIN MARKERBOARD, 60"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- 3 WALL MOUNTED PORCELAIN MARKERBOARD, 72"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- 4 WALL MOUNTED PORCELAIN MARKERBOARD, 84"W X 48"H, MOUNTED AT 5'-0" AFF TO CENTER
- 13 MECHANICAL EQUIPMENT, COORDINATE WITH MECHANICAL DRAWINGS
- 15 EQUIPMENT RACKS (GFGI)
- 17 RAISED ACCESS FLOOR, COORDINATE WITH FINISH LEGEND ON IN602
- 24 ELECTRIC TIME ZONE CLOCK ABOVE
- 34 RACK STORAGE LOCATION, REFER TO AUDIOVISUAL DRAWINGS
- 35 DISPLAY CONTROL LOCATION, REFER TO AUDIOVISUAL DRAWINGS
- 75 CLEAN AGENT TANK, REFER TO FIRE PROTECTION DRAWINGS
- 76 WALL MOUNTED TACKBOARD, 48"W X 36"H, MOUNTED AT 5'-6" AFF TO CENTER, COORDINATE WITH INTERIORS DRAWINGS

KEY PLAN



GRAPHIC SCALE(S)



APPR	DATE	3/1/20
SYN	DESCRIPTION	1. MISS. REVISIONS

COMMONWEALTH OF VIRGINIA
ROSE A. MOSER
Lic. No. 013512
2026.03.10 14:08:37-04'00"
ARCHITECT
CLARK NEXSEN
4525 MAIN STREET, SUITE 1400
VIRGINIA BEACH, VA 23462
757.455.5800
APPROVED: [Signature]
FOR COMMANDER NAVFAC
ACTIVITY: 3/16/2020
Dave Snelling MQ-25 SEIT APML
SATISFACTORY TO DATE: 05/28/2025
DES: RAM | DRW: KCM | CHK: SEB
PM/DIM: RY / JFG
BRANCH MANAGER: DJD
CHIEF ENGINEER: JMW
FIRE PROTECTION: HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
P-222 MQ-25 AIRCRAFT LAYDOWN FACILITIES,
BUILDING LP-48A AND GBDA TOWER
SCALE: AS NOTED
EPROJECT NO.: 1737012
CONSTR. CONTR. NO.:
NAVFAC DRAWING NO.: 12924207
SHEET 128 OF 400
AE102B
DRAWING REVISION: 26 AUGUST 2020

SECOND FLOOR PLAN - AREA B

1

2

3

UNCLASSIFIED

4

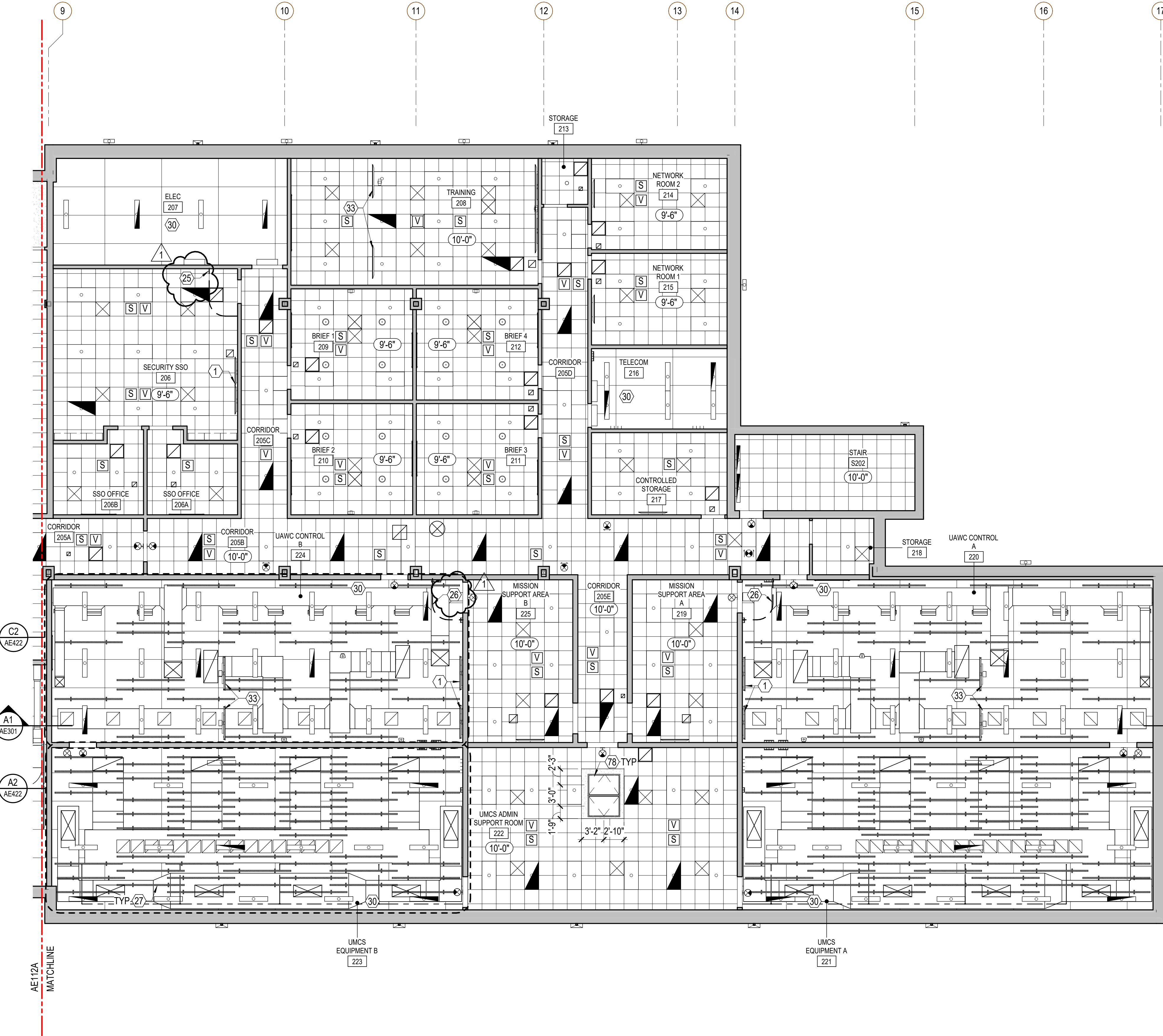
5

D

C

B

A



A1 SECOND FLOOR RCP - AREA B
1/8" = 1'-0"

GENERAL NOTES

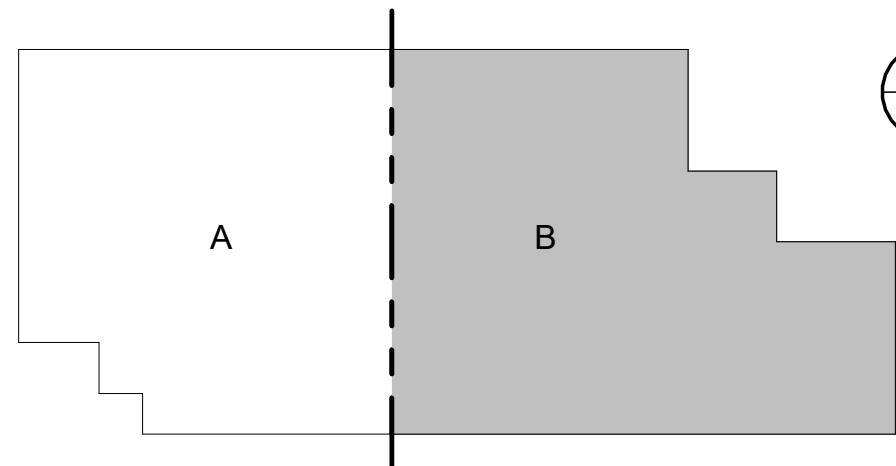
- A. REFER TO SHEET A-001 FOR RCP LEGEND
- B. REFER TO SHEET AE507 FOR CEILING DETAILS.
- C. ALL CEILINGS ARE 9'-0" AFF, UON.
- D. FIELD VERIFY CEILING HEIGHTS SHOWN ON RCP DRAWINGS PRIOR TO INSTALLATION OF ALL MECHANICAL DUCTWORK, SPRINKLER PIPING, ELECTRICAL EQUIPMENT, CONDUITS, ETC. NOTIFY ARCHITECT IMMEDIATELY OF ANY CONFLICTS OR DISCREPANCIES.
- E. ACOUSTICAL TILE CEILING MUST BE CENTERED IN EACH ROOM AS SHOWN ON DRAWINGS, UON.
- F. HVAC & LIGHTING FIXTURES SHOWN ARE FOR GENERAL COORDINATION PURPOSES ONLY. REFER TO PME DRAWINGS AND SPRINKLER PIPING SHOP DRAWINGS. NOTIFY ARCHITECT IMMEDIATELY OF ANY CONFLICTS OR DISCREPANCIES.
- G. COORDINATE ALL SOFFIT/ BULKHEAD FRAMING WITH FINAL LIGHTING FIXTURES & HVAC DIFFUSER LOCATIONS.
- H. INSTALL WIRE HANGERS @ ALL CORNERS OF LIGHT FIXTURES, REGISTERS, & GRILLES.
- I. REFER TO INTERIOR & EXTERIOR SECTIONS FOR ADDITIONAL CEILING DETAILS NOT NOTED ON THIS SHEET.
- J. ALL RECESSED LIGHTING MUST BE CENTERED WITHIN THE SOFFIT DIMENSIONS, UON.
- K. ALL FIXTURES INCLUDING BUT NOT LIMITED TO LIGHTING, MOTION DETECTORS, FIRE ALARM DEVICES, & SPRINKLER HEADS MUST BE CENTERED WITHIN THE ACOUSTICAL CEILING TILE, UON.
- L. WHERE GWB SOFFITS SPAN LONGER THAN 30'-0", CONTROL JOINTS SHOULD BE LOCATED IN GWB SOFFIT EVERY 30'-0" MAX. ALIGN WITH CONTROL JOINTS IN GWB WALLS WHERE PRESENT.

KEY NOTES

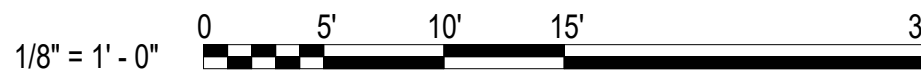
- 1 WALL MOUNTED DISPLAY MONITOR, COORDINATE WITH AUDIOVISUAL DRAWINGS
- 25 PRIVACY CURTAIN, CONTRACTOR TO PROVIDE GRID-MOUNTED TRACK AND MOUNTING SYSTEM
- 26 PRIVACY CURTAIN, CONTRACTOR TO PROVIDE SUSPENDED TRACK AND MOUNTING SYSTEM ON TENSION RODS
- 27 SOUND ABSORBING CEILING BAFFLES, COORDINATE WITH INTERIORS DRAWINGS
- 30 SLOPE RF PANELS TO MATCH SLOPE OF ROOF JOISTS, REFER TO BUILDING SECTIONS
- 33 LOCATION OF CEILING MOUNTED DISPLAY MONITOR (NIC), COORDINATE WITH AUDIOVISUAL DRAWINGS
- 78 36" X 54" ACCESS PANEL, SEE DETAIL A3/AE510

KEY PLAN

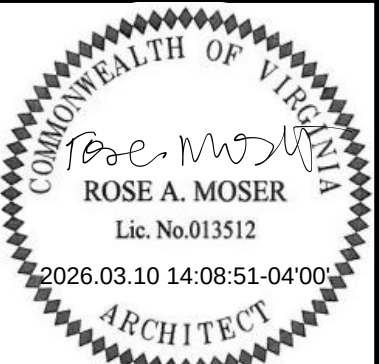
NO SCALE



GRAPHIC SCALE(S)



DATE	SYMBOL	DESCRIPTION
3/1/28	APPR	



CLARK NEXSEN

4525 MAIN STREET, SUITE 1400
VIRGINIA BEACH, VA 23462
757.455.5800

APPROVED	
FOR COMMANDER NAVFAC	3/16/2025
ACTIVITY	Dave Snelling MQ-25 SEIT APML
SATISFACTORY TO DATE	05/28/2025
DES	RAM
DRW	KCM
CHK	SEB
PMIDM	RY / JPC
BRANCH MANAGER	DJD
CHIEF ENGINEER	JWU
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
P-222 MQ-25 AIRCRAFT LAYDOWN FACILITIES,
BUILDING LP-48A AND GBDA TOWER
SECOND FLOOR REFLECTED CEILING PLAN - AREA B

SCALE: AS NOTED
PROJECT NO.: 1737012
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12924214
SHEET 135 OF 400
AE112B
DRAWING REVISION: 26 AUGUST 2020

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2

3

UNCLASSIFIED

4

5



D1

AE312

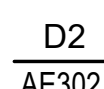


B1
AE311

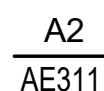


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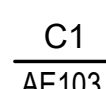


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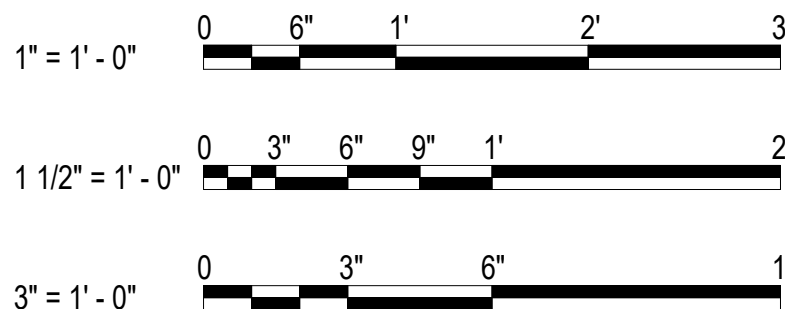


A2

AE311



AE103



FIRE ALARM LEGEND

GENERAL

	DEMOLITION NOTE IDENTIFICATION
	CONSTRUCTION NOTE IDENTIFICATION
---	INDICATES ITEM TO BE REMOVED
	REMOVE TO THIS POINT
	CONNECT NEW TO EXISTING
XXX	ROOM NUMBER IDENTIFICATION
AFF	ABOVE FINISHED FLOOR
PIV	POST INDICATOR VALVE
BFP	BACK FLOW PREVENTER
O.C.	ON CENTER
LH	LIGHT HAZARD
OH	ORDINARY HAZARD
FH	FIRE HYDRANT
EP	ELECTRICAL POWER (120V AC)
EX	ELECTRICALLY CLASSIFIED
WP	WEATHERPROOF (SUBSCRIPT)
C	CEILING MOUNTED (SUBSCRIPT)
E	EXISTING (SUBSCRIPT)
ERCES	EMERGENCY RESPONDER COMMUNICATION ENHANCEMENT SYSTEM
ASSD	AIR SAMPLING SMOKE DETECTION
A.O.R.	AREA OF REFUGE

FIRE ALARM

	FIRE ALARM MASS NOTIFICATION UNIT		MANUAL PULL STATION, MOUNT 4'-0" AFF
	REMOTE FIRE ALARM MASS NOTIFICATION UNIT		STROBE UNIT W/ CANDELA RATING, CEILING MOUNTED
	CLEAN AGENT RELEASING PANEL		STROBE UNIT W/ CANDELA RATING, WALL MOUNTED
	TELEGRAPHIC TRANSMITTER		SPEAKER/STROBE UNIT W/ CANDELA RATING, WALL MOUNTED
	MASS NOTIFICATION RADIO TRANSCEIVER		SPEAKER/STROBE UNIT W/ CANDELA RATING, CEILING MOUNTED
	REMOTE ANNUNCIATOR		SPEAKER, CEILING MOUNTED
	LOCAL OPERATING CONSOLE		EXTERIOR WEATHERPROOF SPEAKER, WALL MOUNTED
	NOTIFICATION APPLIANCE CIRCUIT PANEL		SPEAKER, WALL MOUNTED
	LED TEXT MESSAGING SIGN		CLEAN AGENT PRE-DISCHARGE HORN/STROBE UNIT W/ CANDELA RATING, WALL MOUNTED
	CARBON MONOXIDE DETECTOR, WALL MOUNTED, MOUNT AT 7'-0" AFF		CLEAN AGENT DISCHARGE STROBE UNIT W/ CANDELA RATING, WALL MOUNTED
	DUCT SMOKE DETECTOR		CLEAN AGENT MANUAL RELEASE STATION
	DUCT DETECTOR REMOTE TEST SWITCH		CLEAN AGENT ABORT BUTTON
	SMOKE DETECTOR, CEILING MOUNTED		
	SMOKE DETECTOR, WALL MOUNTED		
	WATERFLOW SWITCH - PADDLE TYPE		
	CONTACT MONITOR MODULE		
	TAMPER SWITCH		
	CONTROL INTERFACE		
	SURGE SUPPRESSOR		

RATED PARTITION LEGEND

ONE HOUR FIRE RATING

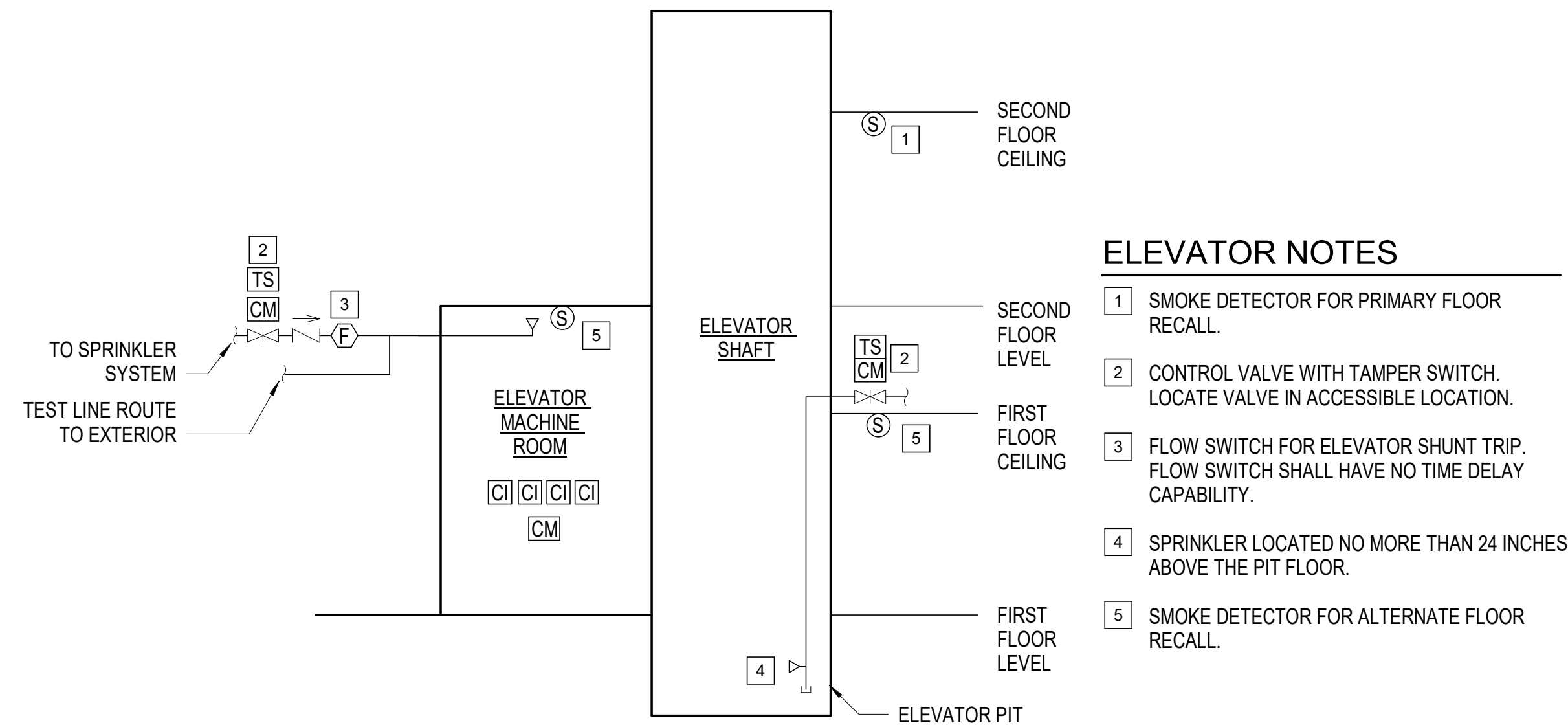
CONTROLLED BOUNDARY LEGEND

RF BOUNDARY
CONTROLLED AREA BOUNDARY

RADIO TRANSMITTER ZONES

ZONE	SIGNAL DESCRIPTION
1	GENERAL ALARM SIGNAL
2	GENERAL TROUBLE SIGNAL
3	GENERAL SUPERVISORY SIGNAL
4	ASSD/CLEAN AGENT ALARM SIGNAL
5	ASSD/CLEAN AGENT TROUBLE SIGNAL
6	ASSD/CLEAN AGENT SUPERVISORY SIGNAL
7	CO DETECTOR
8	DUCT SMOKE DETECTORS

B



ELEVATOR NOTES

- SMOKE DETECTOR FOR PRIMARY FLOOR RECALL.
- CONTROL VALVE WITH TAMPER SWITCH. LOCATE VALVE IN ACCESSIBLE LOCATION.
- FLOW SWITCH FOR ELEVATOR SHUNT TRIP. FLOW SWITCH SHALL HAVE NO TIME DELAY CAPABILITY.
- SPRINKLER LOCATED NO MORE THAN 24 INCHES ABOVE THE PIT FLOOR.
- SMOKE DETECTOR FOR ALTERNATE FLOOR RECALL.

ELEVATOR FIRE PROTECTION DIAGRAM

SCALE: 1/8" = 1'-0"

GENERAL FIRE ALARM/MASS NOTIFICATION SYSTEM NOTES

- PROVIDE AN ANALOG/ADDRESSABLE COMBINED FIRE ALARM AND MASS NOTIFICATION SYSTEM IN ACCORDANCE WITH NFPA 72 - NATIONAL FIRE ALARM AND SIGNALING CODE, 2022 EDITION. CONTRACTOR MUST SUBMIT SHOP DRAWINGS FOR REVIEW. COORDINATE FIRE ALARM SYSTEM WITH ALL TRADES TO ENSURE PROPER COMPLIANCE WITH CODES AND TO AVOID CONFLICTS.
- PROVIDE ADDRESSABLE CLASS A SIGNALING LINE CIRCUITS (SLC), CLASS A NOTIFICATION APPLIANCE CIRCUITS (NAC), AND CLASS B INITIATING DEVICE CIRCUITS (IDC).
- EQUIPMENT AND DEVICES MUST BE UL LISTED FOR USE IN FIRE ALARM SYSTEMS.
- FIRE ALARM/ MASS NOTIFICATION CONTROL PANELS AND EQUIPMENT MUST NOT BE INTERNET PROTOCOL (IP) ENABLED.
- LOCATE CONTROL PANELS IN A YEAR-ROUND ENVIRONMENTALLY CONDITIONED SPACE THAT COMPLIES WITH THE PANEL LISTING FOR TEMPERATURE AND HUMIDITY RANGES.
- PROVIDE SEPARATE ENCLOSURES FOR THE FMCP (SECURED), LOC (ACCESSIBLE), REMOTE ANNUNCIATORS (SECURED) AND PUBLIC ADDRESS MICROPHONES (ACCESSIBLE). TO MEET THIS REQUIREMENT, FIELD MODIFICATIONS TO PANEL DOORS MAY ONLY BE MADE:
 - WITH WRITTEN MANUFACTURER CONCURRENCE THAT THE LISTING IS MAINTAINED.
 - WHERE THE NATURE OF THE SPACE WILL NOT INTRODUCE EXCESSIVE DUST OR DEBRIS INTO THE ENCLOSURE, AND
 - WHERE THE SECURED FUNCTIONS CANNOT BE ACCESSED THROUGH OPENINGS BY REACHING INTO GUTTER SPACE.
- CIRCUITS/WIRING MUST BE IN CONDUIT (MINIMUM 3/4" EMT). VERTICAL DROPS DIRECTLY TO A SINGLE DEVICE/APPLIANCE MAY BE 1/2". CONDUITS MUST BE CONCEALED IN WALLS, FLOOR SLABS OR CEILINGS UNLESS OTHERWISE INDICATED. EXCEPTION: CONDUIT MUST BE ALLOWED TO BE SURFACE MOUNTED ON WALLS AND CEILINGS IN UNFINISHED AREAS (E.G., MECHANICAL ROOM, ELECTRICAL ROOM, RISER ROOM), WHERE CONDUIT FEEDS DEVICES NOT MOUNTED DIRECTLY ON THE WALL OR CEILING, THE CONDUIT MUST BE ROUTED AS A SINGLE PENDENT TYPE DROP TO EACH DEVICE.
- FIRE ALARM CONDUITS AND JUNCTION BOXES IN UNFINISHED AREAS (INCLUDING AREAS ABOVE CEILINGS) MUST BE PRE-FINISHED RED FROM THE FACTORY. CONDUIT AND JUNCTION BOXES EXPOSED IN FINISHED AREAS MUST BE PAINTED TO MATCH ADJACENT FINISHES. [DESIGNER NOTE: COORDINATE PAINTING IN SPEC] JUNCTION BOXES MUST HAVE A PERMANENT, MACHINE PRINTED LABEL READING "FIRE ALARM CIRCUIT CONTROL" ON THE INSIDE COVER.
- WIRING, CABLES, BOXES, TROUGHS, RACEWAYS, AND OTHER RELATED EQUIPMENT MUST BE INSTALLED IN ACCORDANCE WITH NFPA 70, NATIONAL ELECTRICAL CODE (NEC) 2023 EDITION. CONDUIT MUST BE SUPPORTED IN STRICT ACCORDANCE WITH THE NEC (E.G., 10 FT INTERVALS AND 3 FT FROM TERMINATION). TENSION ONLY HANGERS ARE NOT PERMITTED (E.G., BATWINGS). TYPE OF CONDUIT UTILIZED MUST BE IN ACCORDANCE WITH THE ELECTRICAL SPECIFICATIONS UNLESS OTHERWISE NOTED. FLEXIBLE CONDUIT IS PROHIBITED FOR NOTIFICATION APPLIANCE CIRCUITS AND SIGNALING LINE CIRCUITS. FLEXIBLE CONDUIT IS ONLY PERMITTED FOR INITIATING DEVICE CIRCUITS 6 FEET MAXIMUM IN LENGTH.
- CONDUCTORS MUST RUN FROM DEVICE TO DEVICE WITHOUT SPLICES. WIRE NUTS AND CRIMPED CONNECTORS ARE PROHIBITED. TERMINAL STRIPS MUST BE UTILIZED WHERE CONDUCTORS CANNOT BE DIRECTLY TERMINATED AT THE DEVICE (E.G., PIGTAILS OFF A HEAT DETECTOR).
- FIRE ALARM EQUIPMENT PANELS MUST NOT BE USED AS RACEWAY FOR ROUTING POWER WIRING OR LOW VOLTAGE WIRING. ONLY WIRING TERMINATING WITHIN THE PANEL MUST BE ROUTED INTO THE PANEL. POWER WIRING AND POWER LIMITED WIRING MUST BE SEPARATED BY A MINIMUM OF TWO INCHES WITHIN THE EQUIPMENT PANELS.
- POWER SUPPLIES TO FACP, NAC BOOSTER PANELS, AMP BOOSTER PANELS, AND TRANSCEIVER PANELS MUST BE PROVIDED WITH SURGE PROTECTION DEVICES (SPD). SLC, NAC, OR IDC ENTERING OR LEAVING THE BUILDING MUST BE PROVIDED WITH SPD. EXCEPTION: CIRCUITS TO DEVICES MOUNTED DIRECTLY ON THE EXTERIOR OF THE BUILDING DO NOT REQUIRE SPD (E.G., EXTERIOR ALARM BELL). SPD MUST BE MOUNTED IN A SEPARATE ENCLOSURE.
- PROVIDE WALL-MOUNTED VISUAL NOTIFICATION APPLIANCES 6 INCHES BELOW FINISHED CEILING OR 80 INCHES AFF, WHICHEVER IS LOWER, TO THE BOTTOM OF THE LENS. PROVIDE WALL-MOUNTED AUDIBLE NOTIFICATION APPLIANCES 80" AFF MINIMUM MEASURED TO THE BOTTOM OF THE DEVICE AND 96" AFF MAXIMUM MEASURED TO THE TOP OF THE DEVICE. MOUNT INTERFACE MODULES, MONITOR MODULES, CONTROL MODULES AND REMOTE TEST SWITCHES NO HIGHER THAN 6 FEET AFF.
- CEILING MOUNTED FA DEVICES AND NOTIFICATION APPLIANCES MUST BE INSTALLED AT THE CENTER POINT OF SUSPENDED CEILING TILE.
- PENETRATIONS OF FIRE RESISTANCE RATED BARRIERS, WALLS, AND SHAFT ENCLOSURES MUST BE CORE-DRILLED AND SEALED WITH APPROVED UL LISTED FIRESTOPPING ASSEMBLIES. COORDINATE WITH FIRESTOPPING CONTRACTOR IN SPECIFICATION DIVISION 7.
- COORDINATE WITH ELECTRICAL CONTRACTOR TO INTERFACE WITH THE LIGHTING CONTROL SYSTEM TO TURN ON DESIGNATED OCCUPANCY SENSOR CONTROLLED MEANS OF EGRESS LIGHTING UPON A FIRE ALARM CONDITION. COORDINATE WITH THE NAVFAC DFPE TO IDENTIFY DESIGNATED MEANS OF EGRESS.
- PROVIDE AUDIBLE NOTIFICATION FOR ALL AREAS AND ROOMS THROUGHOUT THE FACILITY. THE AUDIBLE REQUIREMENTS MUST BE ACCOMPLISHED WITH ALL DOORS, FIRE SHUTTERS, MOVEABLE PARTITIONS, WINDOWS, AND SIMILAR ELEMENTS CLOSED. WHERE THE AUDIBLE REQUIREMENTS ARE NOT ACCOMPLISHED DURING SYSTEM TESTING, PROVIDE ADDITIONAL APPLIANCES TO MEET THE MINIMUM DESIGN CRITERIA REQUIREMENTS AT NO ADDITIONAL COST. AS-BUILT DRAWINGS MUST DEMONSTRATE THAT THE ADDITIONAL DEVICES AND OR MODIFICATIONS TO THE INSTALLED DEVICES DO NOT EXCEED THE LIMITATIONS OF THE FACP, SNAC BOOSTER PANEL, OR AMP BOOSTER PANEL (INCLUSIVE OF SAFETY FACTORS AND REQUIRED SPARE CAPACITY).
- ALTHOUGH AUDIBLE NOTIFICATION APPLIANCES ARE NOT REQUIRED IN EVERY ROOM, SPEAKER PERFORMANCE MUST ADHERE TO THE AUDIBILITY AND INTELLIGIBILITY CRITERIA WITHIN THE CONTRACT DOCUMENTS.
- AUDIBLE NOTIFICATION APPLIANCES (SPEAKERS) ARE NOT REQUIRED IN STAIR ENCLOSURES UNLESS THE FACILITY REQUIRES RELOCATION OR A DEFEND IN PLACE APPROACH.
- PROVIDE EXTERIOR SPEAKERS AT ALL EXTERIOR DOORS. A SINGLE SPEAKER MAY BE PROVIDED WHERE MULTIPLE DOORS ARE IN THE SAME VICINITY.
- PROVIDE WEATHERPROOF NOTIFICATION APPLIANCES IN AREAS OF SHOWERS, DAMP LOCATIONS, DRYING AREAS, POTENTIALLY CORROSIVE ENVIRONMENTS, UNCONDITIONED AREAS IN SALT AIR ENVIRONMENTS, AND AREAS WITH EXTREME TEMPERATURE CONDITIONS.
- THE SOUND PRESSURE LEVEL OF THE AUDIBLE NOTIFICATION MUST BE A MINIMUM OF 70 DBA AND 15 DBA OVER THE AMBIENT CONDITIONS AT THE MOST REMOTE LOCATION WITHIN THE ROOM OR AREA. THE SOUND PRESSURE LEVEL MUST BE MEASURED OFF THE TEMPORAL TONE PROCEEDING EACH VOICE MESSAGE. THE FOLLOWING CRITERIA MUST BE UTILIZED FOR REVIEW AND APPROVAL OF SHOP DRAWINGS.
 - THE SOUND PRESSURE LEVEL MUST BE REDUCED BY 6 DBA EACH TIME THE DISTANCE BETWEEN THE APPLIANCE AND THE LISTENER IS DOUBLED. STARTING DISTANCE IS PER MANUFACTURER'S DATASHEET, TYPICALLY 10 FT.
 - THE SOUND PRESSURE LEVEL MUST BE REDUCED IN ACCORDANCE WITH THE MANUFACTURER'S DATASHEET FOR SOUND PRESSURE LEVEL DISTRIBUTION NOT PERPENDICULAR TO THE SPEAKER FACE. REFER TO MANUFACTURER'S TYPICAL SOUND OUTPUT DISTRIBUTION DIAGRAM.
 - A MINIMUM 15 DBA LOSS MUST BE ASSUMED THROUGH A STANDARD CLOSED DOOR. LOSSES FOR DOORS WITH STC RATINGS GREATER THAN STANDARD DOORS MUST BE CONSIDERED BASED ON THE STC RATING.
- SOUND PRESSURE LEVEL MUST NOT EXCEED 110 DBA IN OCCUPIABLE AREAS.
- INTELLIGIBILITY MUST MEET THE FOLLOWING CRITERIA:
 - NORMALLY UNOCCUPIED ROOMS ARE CONSIDERED SMALL STORAGE ROOMS (< XX SF), MECHANICAL ROOMS, RISER ROOMS, ELECTRICAL ROOMS, JANITOR CLOSETS, AND COMMUNICATION CLOSETS (< XX SF). ALL OTHER ROOMS ARE CONSIDERED NORMALLY OCCUPIED.
 - A CIS SCORE OF 0.70 MUST BE ACHIEVED IN THE MOST REMOTE AREA OF EACH ROOM AT 5 FT ABOVE THE FINISHED FLOOR. WHEN MEASURING INTELLIGIBILITY, ROUNDING OF VALUES IS NOT PERMITTED.
 - AREAS OF THE BUILDING PROVIDED WITH HARD WALL AND CEILING SURFACES (SUCH AS METAL OR CONCRETE) THAT ARE FOUND TO CAUSE EXCESSIVE SOUND REFLECTIONS MAY BE PERMITTED TO HAVE A CIS SCORE LESS THAN THE MINIMUM REQUIRED VALUE IF BUILDING OCCUPANTS IN THESE AREAS CAN DETERMINE THAT A VOICE SIGNAL IS BEING BROADCAST AND THEY MUST WALK NO MORE THAN 33 FT TO A LOCATION WITH AT LEAST THE MINIMUM REQUIRED CIS VALUE WITHIN THE SAME AREA.
 - NORMALLY UNOCCUPIED AREAS OF THE BUILDING ARE PERMITTED TO HAVE A CIS SCORE LESS THAN THE MINIMUM REQUIRED VALUE IF PERSONNEL CAN DETERMINE THAT A VOICE SIGNAL IS BEING BROADCAST AND THEY MUST WALK NO MORE THAN 50 FT TO A LOCATION WITH AT LEAST THE MINIMUM REQUIRED CIS VALUE WITHIN THE SAME AREA.
- STROBE SPACING FOR THE FIRE ALARM NOTIFICATION SYSTEM MUST BE IN ACCORDANCE WITH NFPA 72. IN AREAS WITH CEILINGS, PROVIDE CEILING-MOUNTED AUDIOVISUAL NOTIFICATION APPLIANCES. IN AREAS WITHOUT CEILINGS, PROVIDE SUSPENDED OR WALL-MOUNTED AUDIOVISUAL NOTIFICATION APPLIANCES.
- PROVIDE VISUAL NOTIFICATION APPLIANCES IN ALL NORMALLY OCCUPIED, PUBLIC AND COMMON USE AREAS. PROVIDE VISUAL NOTIFICATION APPLIANCE IN NORMALLY UNOCCUPIED AREAS (SUCH AS MECHANICAL ROOMS, ELECTRICAL ROOMS, JANITOR ROOMS, STORAGE AREAS, COMMUNICATION CLOSETS OR OTHER SIMILAR SPACES) GREATER THAN 900 SQ.FT.
- PROVIDE VISUAL NOTIFICATION IN NORMALLY UNOCCUPIED AREAS WHERE THE AMBIENT NOISE IS LOUD ENOUGH TO REQUIRE HEARING PROTECTION.
- PROVIDE VISUAL NOTIFICATION IN OFFICES THAT ARE DESIGNED FOR, OR MAY CONTAIN, MORE THAN FOUR PERSONS AT ANY ONE TIME. THIS IS DETERMINED BY THE FURNITURE PLAN AND CHAIRS WITH WORK SURFACES.
- CARBON MONOXIDE DETECTION MUST COMPLY WITH NFPA 720. CARBON MONOXIDE DETECTORS MUST BE INSTALLED AFTER CLEANUP OF ALL OTHER CONSTRUCTION TRADES.
- MULTI-CRITERIA DETECTORS ARE PROHIBITED.
- MANUAL STATIONS MUST BE PROVIDED AT ALL EXITS. THE 200 FOOT TRAVEL DISTANCE REQUIRED BY NFPA 101 DOES NOT APPLY.
- PROVIDE A SEPARATE ADDRESS FOR EACH SPRINKLER WATERFLOW SWITCH.
- NO MORE THAN FIVE ADJACENT VALVE TAMPER SWITCHES WITHIN THE SAME ROOM MAY BE MONITORED BY THE SAME ADDRESSABLE DEVICE OR SUPERVISORY CIRCUIT.
- THE FIRE ALARM DRAWINGS ARE SCHEMATIC IN NATURE AND SHOW A MINIMAL QUANTITY OF DEVICES. THEY DO NOT SHOW THE EXACT LOCATION OF COMPONENTS NOR SHOW ALL SYSTEM COMPONENTS. CONTRACTOR MUST DETERMINE THE FINAL QUANTITY AND LOCATION OF DEVICES/COMPONENTS IN ACCORDANCE WITH THE SPECIFICATIONS, CONTRACT DRAWINGS, AND MANUFACTURER'S WRITTEN RECOMMENDATIONS TO BE COMPLIANT WITH ALL APPLICABLE CODES AND STANDARDS. PROVIDE ADDITIONAL DEVICES FOR A FULLY FUNCTIONING FIRE ALARM SYSTEM, SUBJECT TO APPROVAL BY THE NAVFAC DESIGNATED FIRE PROTECTION ENGINEER (DFPE).
- REFER TO SHEET EH101 FOR LOCATION OF ELECTRICALLY CLASSIFIED SPACES. ALL FIRE ALARM COMPONENTS AND INSTALLATION IN THESE AREAS MUST BE SUITABLE FOR THE INDICATED CLASSIFICATIONS.
- REFER TO B1/GI202 FOR SECURE BOUNDARY PENETRATION DETAIL WITH FIRE ALARM CONDUIT.

3/1/26

DATE

APPR

1

MISC REVISIONS

SYM DESCRIPTION

COMMONWEALTH OF VIRGINIA

CHRISTOPHER H. BORN

Lic. No. 23010

PROFESSIONAL ENGINEER

CLARK NEXSEN

AVE INFO

4525 MAIN STREET, SUITE 1400

VIRGINIA BEACH, VA 23462

757.455.5800

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

Dave Snelling MQ-25 SEIT APML

SATISFACTORY TO DATE

05/28/2025

DES TRJ DRW TRJ CHK CHB

PMIDM RYJPG

BRANCH MANAGER HPN

CHIEF ENGINEER JWWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

P-222 MQ-25 AIRCRAFT LAYDOWN FACILITIES,

BUILDING LP-48A AND GBDA TOWER

FIRE ALARM LEGEND

SCALE: AS NOTED

EPROJECT NO.: 1737012

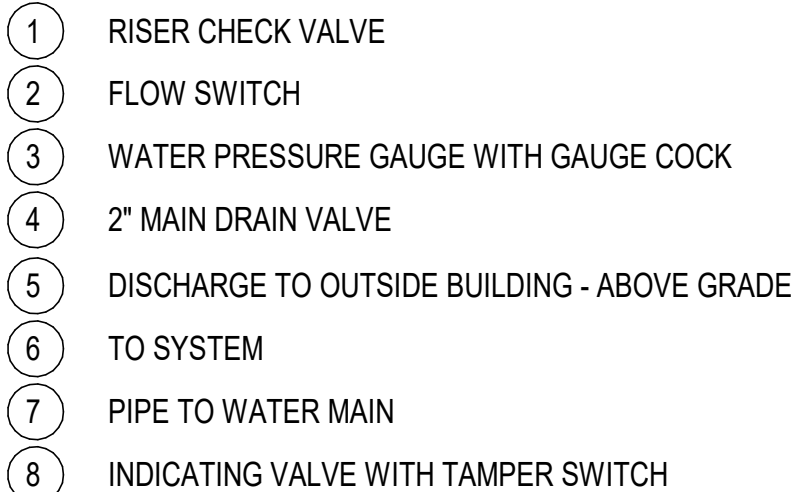
CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12924284

SHEET 205 OF 400

FA001

DRAWING REVISION: 25 AUGUST 2020



NOT TO SCALE



3. PROVIDE A COMPLETE HYDRAULICALLY DESIGNED AUTOMATIC WET-PIPE SPRINKLER SYSTEM IN ACCORDANCE WITH NFPA 13, 2022 EDITION FOR COMPLETE COVERAGE THROUGHOUT THE BUILDING. THE SYSTEM MUST INCLUDE QUICK RESPONSE SPRINKLERS, WATER FLOW SWITCHES, TAMPER SWITCHES, CHECK VALVES, CONTROL VALVES, PIPING, HANGERS AND ALL ASSOCIATED EQUIPMENT. WORK AND INSTALLATION MUST CONFORM WITH APPLICABLE PROVISIONS OF THE CODES AND STANDARDS LISTED IN THE CONTRACT DOCUMENTS.
2. COORDINATION BETWEEN TRADES MUST BE TO THE SATISFACTION OF THE DPFE AND CONTRACTING OFFICER. ANY DEFICIENCIES, INCONSISTENCIES, OR POORLY COORDINATED INSTALLATIONS MUST BE CORRECTED BY THE CONTRACTOR AT NO ADDITIONAL COST.
3. RUN PIPING IN ROOMS AND CORRIDORS CONCEALED ABOVE CEILINGS. IN ROOMS WITH CEILINGS OF EXPOSED CONSTRUCTION, RUN THE PIPING EXPOSED USING UPRIGHT SPRINKLER HEADS, EXCEPT AS NOTED OTHERWISE.
4. NO PIPE PENETRATIONS OF STRUCTURAL MEMBERS, EXCEPT AS NOTED, ARE PERMITTED WITHOUT THE APPROVAL OF THE ARCHITECT/ENGINEER.
5. PROVIDE PIPE PENETRATION OF FIRE OR SMOKE PARTITIONS OR WALLS AND MAKE FIRE AND SMOKE TIGHT. PROVIDE UL LISTED THROUGH PENETRATION FIRE STOPPING ASSEMBLIES FOR EACH PENETRATION OF FIRE-RATED ASSEMBLIES.
6. FIRE SPRINKLER SYSTEM, EQUIPMENT, AND COMPONENTS MUST BE UL LISTED OR FM APPROVED FOR FIRE PROTECTION APPLICATION.
7. WET-PIPE FIRE SPRINKLER SYSTEM MUST USE UL LISTED BLACK STEEL PIPE; MINIMUM SCHEDULE 40 FOR PIPING 2-IN AND LESS, AND MINIMUM SCHEDULE 10 FOR PIPING LARGER THAN 2-IN.
8. SPRINKLERS MUST BE INSTALLED AT THE CENTER POINT OF SUSPENDED ACOUSTICAL CEILING TILE.
9. FIRE SPRINKLER PIPE AND FITTINGS MUST BE PAINTED IN RED AND LABELED. AT THE MINIMUM, FEED AND CROSS MAINS MUST BE LABELED TO IDENTIFY THE PIPE FUNCTION; "SPRINKLER SYSTEM," "FIRE DEPARTMENT CONNECTION" WITH ARROW SHOWING WATERFLOW DIRECTION. FOR PIPE SIZES 4-IN AND LARGER, PROVIDE WHITE PAINTED STENCILED LETTERS AND ARROW, A MINIMUM OF 2-IN IN HEIGHT AND VISIBLE FROM AT LEAST 2 SIDES VIEWING FROM THE FLOOR. FOR PIPE SIZES LESS THAN 4-IN, PROVIDE WHITE STENCILED LETTERS AND ARROWS, A MINIMUM OF 0.75-IN IN HEIGHT AND VISIBLE FROM THE FLOOR. CONTROL VALVES MUST BE MARKED WITH PERMANENT TAGS INDICATING "NORMALLY OPEN" OR "NORMALLY CLOSED".
10. THE PROJECT IS A SEISMIC DESIGN CATEGORY A. SEISMIC RESTRAINT IN ACCORDANCE WITH NFPA 13 IS NOT REQUIRED.
11. BACKFLOW PREVENTER TEST HEADERS MUST BE PROVIDED ON THE FACILITY EXTERIOR EXCEPT WHERE THE UNIT IS LOCATED IN A DEDICATED SPRINKLER RISER ROOM WITH DIRECT ACCESS TO THE EXTERIOR AND WHERE HOSES WILL NOT OBSTRUCT EGRESS. CREATE TRIP HAZARDS, OR RESULT IN POTENTIAL LEAKAGE OR DAMAGE TO INTERIOR FINISHES AND EQUIPMENT.
12. FIRE DEPARTMENT CONNECTIONS MUST BE INSTALLED APP. 36" ABOVE GRADE +/- 6 INCHES.
13. ALL DRAIN PENETRATIONS THROUGH EXTERIOR WALLS MUST BE NO GREATER THAN 2'-0" ABOVE FINISHED GRADE AND PROVIDED WITH A SPLASH BLOCK. PROVIDE INSPECTOR'S TEST/DRAIN IN LOCATIONS INDICATED ON THE DRAWINGS AND AS APPROVED BY THE NAVFAC DPFE.
14. FIRE SPRINKLER PIPING MUST NOT PASS THROUGH OR OVER ELECTRICAL ROOMS AND TELECOM ROOMS UNLESS THE PIPING SPECIFICALLY SERVES ONLY SPRINKLERS IN THOSE AREAS. SPRINKLER PIPE MUST BE INSTALLED AS FAR AWAY FROM ELECTRONIC EQUIPMENT AS POSSIBLE AND IN NO CASE WITHIN THE DEDICATED EQUIPMENT SPACE AS DEFINED BY NFPA 70.
15. PROVIDE CORROSION RESISTANT SPRINKLERS IN AREAS OF SHOWERS, DAMP LOCATIONS, DRYING AREAS, POTENTIALLY CORROSIVE ENVIRONMENTS, UNCONDITIONED AREAS IN SALT AIR ENVIRONMENTS, AND AREAS WITH EXTREME TEMPERATURE CONDITIONS.
16. THE FIRE SUPPRESSION SYSTEM MUST BE SERVED FROM THE EXISTING INSTALLATION WATER SUPPLY SYSTEM. PERFORM A HYDRANT FLOW TEST UNDER DIRECT SUPERVISION OF THE DPFE, AND INCORPORATE THIS TEST INTO THE HYDRAULIC CALCULATIONS UNLESS VERIFIED BY THE NAVFAC DPFE AND APPROVED BY THE CONTRACTING OFFICER. HYDRANT FLOW TESTING MUST BE IN ACCORDANCE WITH NFPA 291, AND THE RESULTS OF THE TESTING (HYDRANT FLOW TEST REPORT) MUST BE SUBMITTED IN ACCORDANCE WITH NFPA 291.
17. HYDRAULIC CALCULATIONS MUST INCLUDE A MINIMUM OF 12 PSI PRESSURE DROP ACROSS THE REDUCED PRESSURE (RPZ) BACKFLOW PREVENTER.
18. PROVIDE QUICK-RESPONSE SPRINKLERS IN ALL SPACES FOR WET-PIPE FIRE SPRINKLER SYSTEM. DESIGN AREA REDUCTIONS FOR QUICK-RESPONSE SPRINKLERS ARE NOT PERMITTED.
19. THE USE OF EXTENDED COVERAGE SPRINKLERS IS NOT PERMITTED UNLESS APPROVED BY THE NAVFAC DPFE.
20. PIPING MATERIAL PRIOR TO THE BACKFLOW PREVENTER MUST BE SUITABLE FOR USE IN POTABLE WATER SYSTEMS IN COMPLIANCE WITH THE INTERNATIONAL PLUMBING CODE & LOCAL JURISDICTION.
21. FIRE SPRINKLER DRAWINGS DEMONSTRATE THE CONFIGURATION OF MAJOR SYSTEM COMPONENTS AND SHOW THE DESIGN INTENT. THE DRAWINGS ARE DIAGRAMMATIC IN NATURE AND ARE NOT INTENDED TO SHOW EXACT LOCATIONS AND/OR ASSOCIATED COMPONENTS, PIPING, HANGERS, ETC. PIPE LENGTH AND ELEVATION INDICATED ON DRAWINGS (IF SHOWN) ARE APPROXIMATE. PROVIDE ADDITIONAL COMPONENTS AND EQUIPMENT AS REQUIRED FOR A FULLY FUNCTIONING FIRE SUPPRESSION SYSTEM IN ACCORDANCE WITH NFPA 13. THE CONTRACTOR MUST COORDINATE THE FINAL INSTALLATION WITH ACTUAL FIELD CONDITIONS AND OTHER CONSTRUCTION TRADES.
22. TERMINATE ALL DRAINAGE AND TEST PIPING TO THE EXTERIOR OR FLOOR DRAIN SO IT WILL NOT CAUSE DAMAGE. PROVIDE SPLASH BLOCKS LARGE ENOUGH TO MITIGATE EROSION. ENSURE ALL DISCHARGED WATER DRAINS AWAY FROM THE FACILITY AND DOES NOT CAUSE PROPERTY DAMAGE OR CROSS OR INTERFERE WITH MEANS OF EGRESS PATHS.